

QXU5017 EXP2-2 Report Marking Criteria

Criteria	Excellent (80-100%)	Good (65-79%)	Fair (50-64%)	Needs improvement (0-49%)	Points
Introduction	• Clear and concise aims and objectives. • Accurately identify the main tasks in this experiment. • Justifies the advantages over a tensile test with strong scientific reasoning.	• Clear aims and objectives. • Identify the main tasks in this experiment. • Justifies the advantages over a tensile test but may lack supporting evidence.	• Aims and objectives are vague or not fully relevant. • Describe the tasks in this experiment with minor omissions. • Offers a weak comparison with the tensile test.	• Unclear aims and objectives. • Offers little or no comparison with the tensile test.	12
Method	• Clearly refers to the “Experiment Plan’ without unnecessary repetition. • Provides a concise summary of the key method.	• Refers to the “Experimental Plan” but may include some unnecessary details. • Summarizes the methods adequately but may lack conciseness.	• Mentions the “Experimental Plan” but lacks clarity or include detailed procedure. • Some key aspects of the methods remains unclear.	• Fail to refer to the “Experimental Plan” and instead describes the procedure unnecessarily.	3
Results	• Results are clearly presented using well-labeled tables or figures. • Includes all 3 required plots • Correctly calculate the uncertainty with relevant equations (necessary derivations) . • Final results are presented in proper scientific format with appropriate significant figures. • Complete the uncertainty budget table with all sources of errors. • Properly refers to the Excel file for detailed calculations.	• Results are mostly well-organized using tables and figures with minor inconsistencies. • Includes all 3 required plots • Calculate the uncertainty with essential equations but may lack some clarity • Final results are in scientific format but significant figures could be better rounded. • Complete the uncertainty budget table with key sources of errors. • Refers to the Excel file, but not seamlessly.	• Results are presented but may have some missing or incorrect elements in tables, figures, or labels. • Includes 3 plots but may lack clarity or proper formatting. • Uncertainty calculations are present but not well-explained. • Final results may not be in proper scientific notation. • Uncertainty budget table may not fully reflect the contributions. • The reference to the Excel file is unclear.	• Results are poorly organized or missing key elements in tables or figures. • Uncertainty calculations are incomplete or incorrect. • Final results are not in scientific notation. • Uncertainty budges table is missing. • Does not properly reference the Excel file for detailed calculations.	40

Criteria	Excellent (80-100%)	Good (65-79%)	Fair (50-64%)	Needs improvement (0-49%)	Points
Discussion	- Provides a thorough analysis of the results with clear evaluation of the measurement precision. - Accurately identifies main sources of uncertainty with reference to the uncertainty budget table and suggests realistic improvements. - Gives thoughtful discussion on the accuracy of results based on the known properties of the materials.	- Provides a general analysis of the results with some reference to precision. - Identifies major sources of uncertainty and suggests reasonable improvements. - Good discussion on the accuracy of results based on the known properties of the materials with less detail or depth.	- Provides a basic discussion of measurement precision. - Mentions sources of uncertainty with limited connection to the uncertainty budget table. - Give a general discussion on the results accuracy but lacks supporting from the known properties of the materials.	- Unclear discussion of the measurement precision. - Uncertainty sources are missing or incorrect. - Fails to comment on results accuracy and no attempt to refers to the known properties of the materials.	25
Conclusion	- Clearly and concisely summarises what was learned from the results. - Makes logical conclusions supported by data. - Reflects on the meaning or implications of the results with insight.	- Summarises the main findings accurately. - Draws reasonable conclusions based on the results, though some points may lack depth or clarity.	- Provides a basic summary of the findings. - Conclusions are somewhat related to the results but may be vague or general.	- Fails to clearly summarise the findings. - Conclusions are weak, unsupported, or not clearly linked to the results.	10
Reflection	- Provide an thoughtful and honest reflection on what was learned about laboratory measurement. - Demonstrates personal insight and growth, highlighting specific skills or understandings gained through the experiment.	- Provide a clear reflection on what was learned but lacking depth. - Identifies some key lessons or skills related to measurement, though may lack depth or specific examples.	- Provide basic reflection with few insights on what was learned. - Mentions learning about measurement but lacks personal insight or specific takeaways.	- Show little or no reflection on learning with general thoughts. - Personal reflection is vague, superficial, or missing.	10
Quality of English and Reference	- Exceptionally well-written, no errors, and perfectly formatted references. - Filename, report title, spreadsheet title, authors are clear and correct.	- Well-written with minor grammatical or referencing errors. - Filename, report title, spreadsheet title, authors are mostly correct.	- Acceptable writing with noticeable grammatical errors or referencing issues. - Filename, report title, spreadsheet title, or authors are incorrect or missing.	- Poor writing, frequent errors, and inadequate or incorrect references. - Filename, report title, spreadsheet title or authors is missing.	0